



# **SC2001/CE2101/CZ2101: Algorithm Design & Analysis**

## **Course Introduction**

Asst. Prof. LUO Siqiang

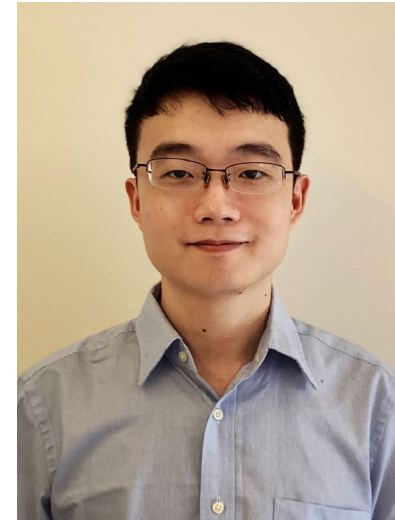
# Instructors

---



**LIU Ziwei, PhD**  
[ziwei.liu@ntu.edu.sg](mailto:ziwei.liu@ntu.edu.sg)

**Course Coordinator**

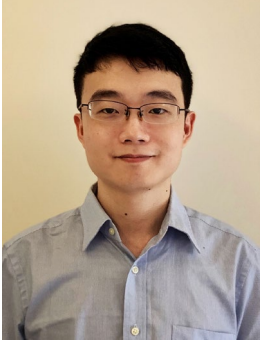


**LUO Siqiang, PhD**  
[siqiang.luo@ntu.edu.sg](mailto:siqiang.luo@ntu.edu.sg)

Consultation Hours: *By email appointment*

# Syllabus

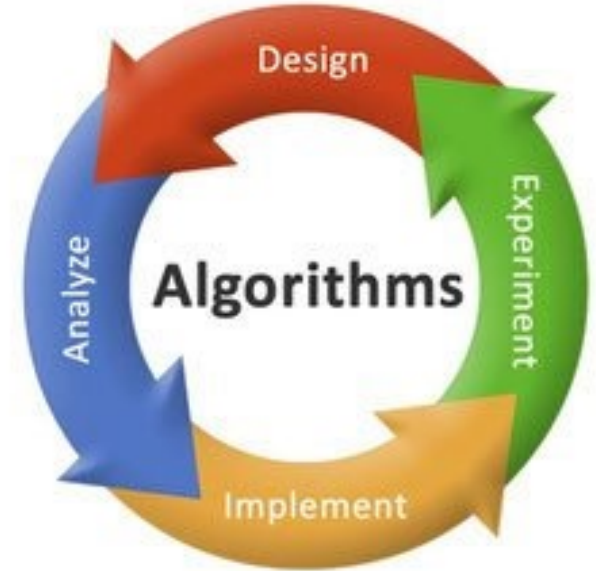
---



- 1. *Sorting (Weeks 1-4)*
- 2. *Graphs (Weeks 4-6)*



- 3. *Analysis Techniques (Weeks 7-8)*
- 4. *Dynamic Programming (Weeks 8-10)*
- 5. *String Matching (Weeks 11-12)*
- 6. *Introduction to NP Completeness (Weeks 12-13)*

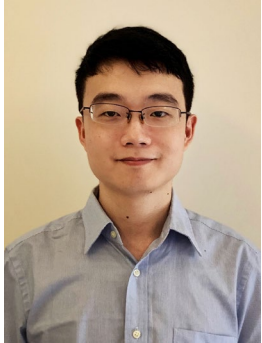


# Learning Objectives & Outcomes

---

- Apply, design and analyse algorithms for
  - sorting, graph problems (Weeks 1-6)
  - string matching, and certain types of optimization problems (Weeks 7-13)
- Upon completion of the course, the student should be able to:
  - a) Design algorithms using suitable strategies (e.g., **incremental, divide-and-conquer, greedy, dynamic programming**) to solve a problem.
  - b) Analyse the **efficiency** of different algorithms for problems studied.
  - c) Understand **complexity** classes P & NP; apply greedy approach to solve NP-complete problems.
  - d) Implement algorithms from pseudo code into real code.

# Learning Plan



| Week        | Lecture Topic                                    | Tutorial                 | Example Classes |
|-------------|--------------------------------------------------|--------------------------|-----------------|
| 1           | Introduction & Sorting (Insertion Sort)          |                          |                 |
| 2           | Sorting (Mergesort)                              |                          |                 |
| 3           | Sorting (Quicksort & Heapsort)                   | Sorting (T1)             |                 |
| 4           | Sorting (Heapsort) & Graphs (Greedy strategy)    | Sorting                  |                 |
| 5           | Graphs (Dijkstra's algorithm & Prim's algorithm) | Sorting                  |                 |
| 6           | Graphs (Union-find & Kruskal's algorithm)        | Graphs (T2)              | Project 1       |
| 7           | Analysis techniques                              | Graphs                   | Project 1       |
| Recess Week |                                                  |                          |                 |
| 8           | Analysis techniques, Dynamic programming         | Graphs                   | Project 2       |
| 9           | Dynamic programming                              | Analysis Techniques (T3) | Project 2       |
| 10          | Dynamic programming                              | Dynamic Programming (T4) | Quiz            |
| 11          | String Matching                                  | Dynamic Programming      | Quiz            |
| 12          | String Matching, Introduction to NP              | String Matching (T5)     | Project 3       |
| 13          | Introduction to NP                               | Introduction to NP (T6)  | Project 3       |

# ***Lecturing Mode (Week 1-13)***

---

- *Lectures: 2 hours per week*
  - *Tuesday, 12:30-13:30*
  - *Thursday, 14:30-15:30*
- *Venue: LT2A*

# Assessments

---

| Assessment                            | Percentage |
|---------------------------------------|------------|
| <i>Example Classes (Team Project)</i> | 20%        |
| <i>Quiz (Individual)</i>              | 20%        |
| <i>Final Exam (Individual)</i>        | 60%        |



# References

- ❑ *Computer Algorithms: Introduction to Design and Analysis.* Sara Baase & Allen Van Gelder, 2000. Third Edition. Addison Wesley. ISBN-10: 0-201-61244-5
- ❑ *Introduction to Algorithms.* Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and C. Stein, 2009. Third Edition. The MIT Press.
- ❑ *Algorithm Design.* J. Kleinberg and E. Tardos, 2005. Addison Wesley.

